



A COMPREHENSIVE EVALUATION OF DEEP LEARNING MODELS FOR CANCER DETECTION ACROSS DIVERSE

Goal: Identify which deep learning model performs best across all cancer types. Evaluate 5 major deep learning architectures: CNN, ResNet, VGG, GoogLeNet, EfficientNet-B0

MOTIVATION

- Interpretation is time-intensive
- Cancer regions are difficult to see due to subtle boundaries and noise
- Early detection is critical
- Manual interpretation can delay diagnosis
- Deep learning offers a scalable approach
- Model performance varies by cancer type
- Cross-cancer evaluation is needed

RESEARCH OBJECTIVE

- Same preprocessing, splits, and metrics for fairness
- Evaluated on 8 cancer datasets
- Test accuracy, Macro-F1, balanced accuracy
- Confusion matrices, ROC, and PR curves

DATASET OVERVIEW

Cancer Types:

- Leukemia, Brain, Breast, Oral, Lymphoma, Cervical, Kidney, Lung and Colon
- All images resized to 224x224
- Pre-split CSVs used for train/val/test to prevent leakage
- 8 main cancer classes 26 subclasses

DATA PIPELINE & PREPROCESSING

- Load image paths from CSV manifest
- Decode images using TensorFlow pipeline
- Resize images to 224x224
- Normalize inputs using Rescaling (1/255)
- Prevent data leakage (no augmentation on validation/test sets)
- Use HPC with NVIDIA 2080Ti GPUs
- Python libraries for preprocessing and training

OVERALL MODEL PERFORMANCE

Model Scores

- EfficientNet-B0: 0.9779, CNN: 0.9015, ResNet: 0.8831, GoogLeNet: 0.6339, VGG: 0.2708

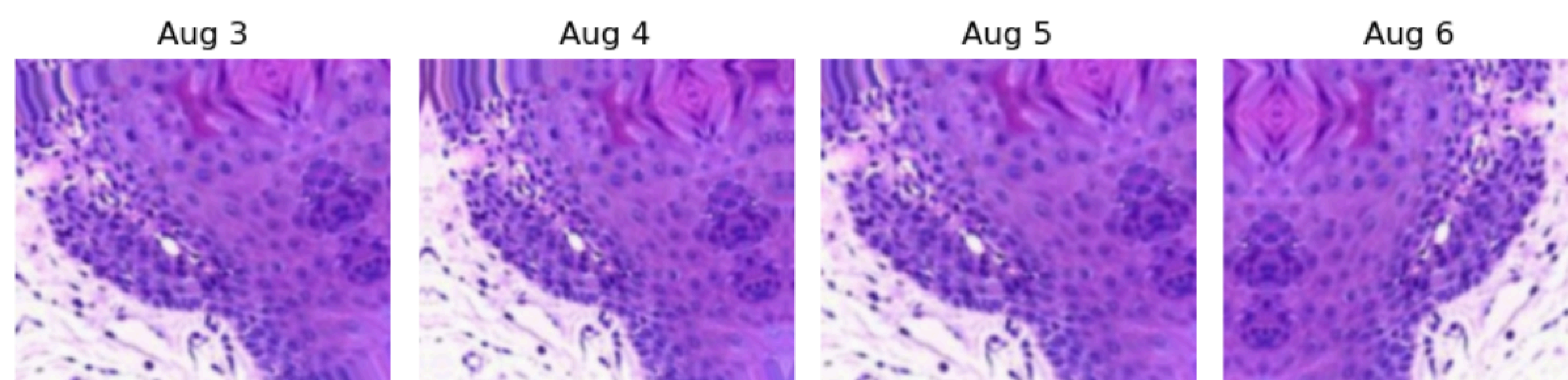
DATA AUGMENTATION

Performed using Keras:

- Performed using Keras:
- Random horizontal flips
- Random rotations
- Random zoom (up to 10%)

Purpose:

- Reduce overfitting
- Improve generalization and performance on unseen cancer images



MODEL ARCHITECTURES EVALUATED

Models:

- CNN, ResNet, VGG, GoogleNet, EfficientNet-B0

Why these architectures?

- Span classical to modern DL models
- Vary in depth, patterns, and scaling
- Offer diverse feature extraction
- Allow fair comparison across model families

TRAINING CONFIGURATION

- Optimizer: Adam (1e-3) | Loss: Sparse Categorical Cross-Entropy
- Batch size: 8 | Epochs: up to 20
- Regularization: EarlyStopping (patience=4, restores best weights)
- Same images, splits, preprocessing for all models
- Compute: HPC with NVIDIA 2080 Ti GPUs | TensorFlow backend

CONCLUSIONS

- CNN: solid but limited depth, less consistent across cancers
- EfficientNet: balanced with attention, better generalization

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ROC & PRECISION - RECALL

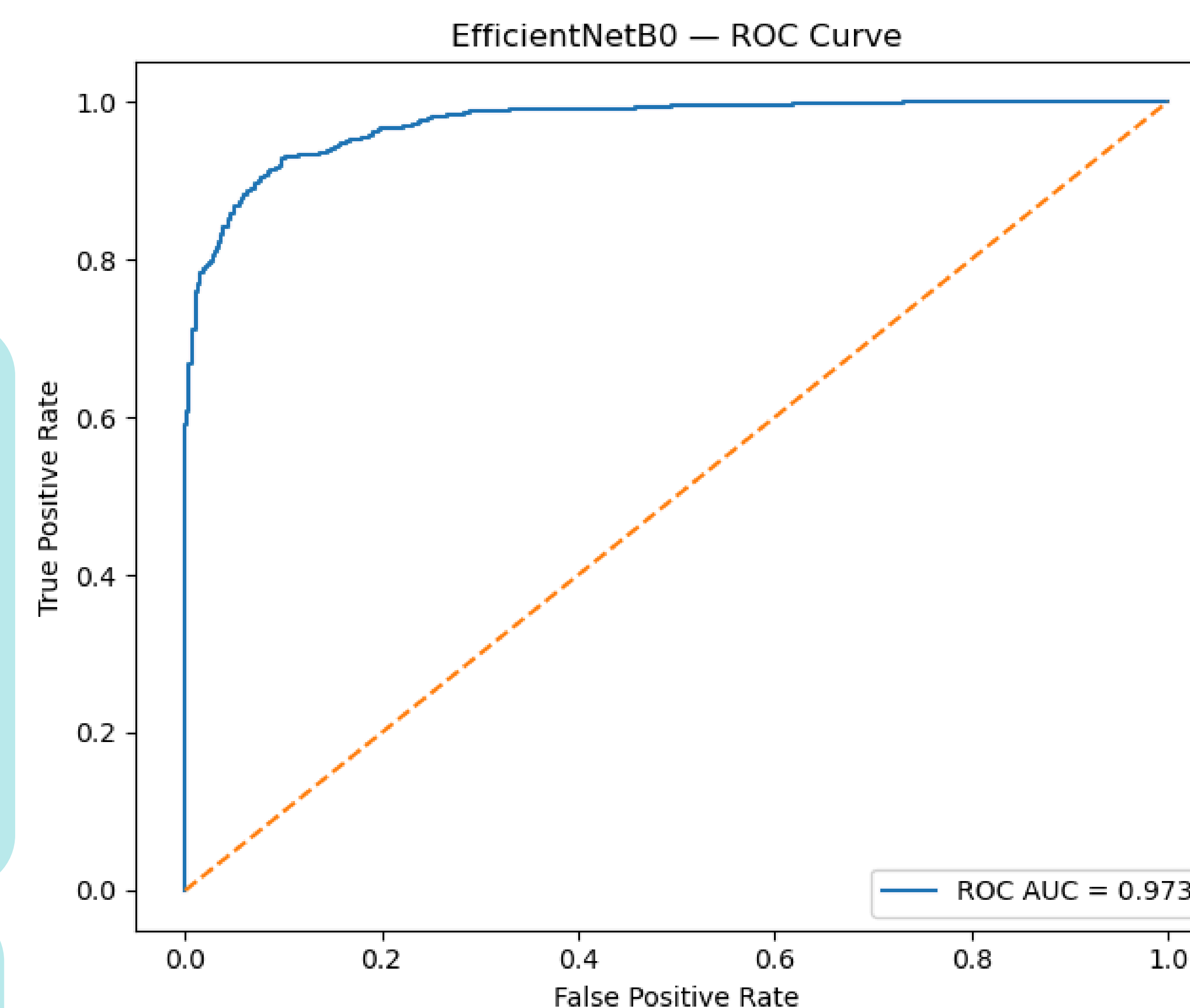
ANALYSIS

Findings:

- ROC performance: AUC ≈ 1.0 for most cancers
- Strong worst-case performance: Lowest AUC (Oral Cancer) ≈ 0.973
- Robust under class imbalance: PR curves remain high, reliably detecting rare positives

Takeaway:

- EfficientNet-B0 is consistently discriminative across thresholds
- Strong PR curves show high precision and recall, not just accuracy



EVALUATION METRICS

Primary Metrics

- Macro-F1 Score
- Best for imbalance medical datasets
- Balanced Accuracy
- Equal weight to each class

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